

Week 3

1. Integrated the official BullMQ Python package and established a Redis-backed queue connection.
2. Implemented a worker to consume jobs from the generate-ai-images queue with an asyncio event loop.
3. Identified duplicate event issue caused by default lockDuration (30s) being shorter than AI job runtime (~50s).
4. Resolved duplication by:
 - Increasing lockDuration to 300000ms (5 minutes)
 - Setting concurrency = 1
 - Enforcing explicit job return values
5. Implemented a worker instance detection utility to prevent multiple concurrent workers from running accidentally.
6. Tested job lifecycle extensively to confirm:
 - Single consumption
 - No reprocessing
 - No duplicate status emissions
7. Documented root cause analysis and resolution strategy for duplicate processing issue.